Lab-Setups – Amazon ECS using Fargate

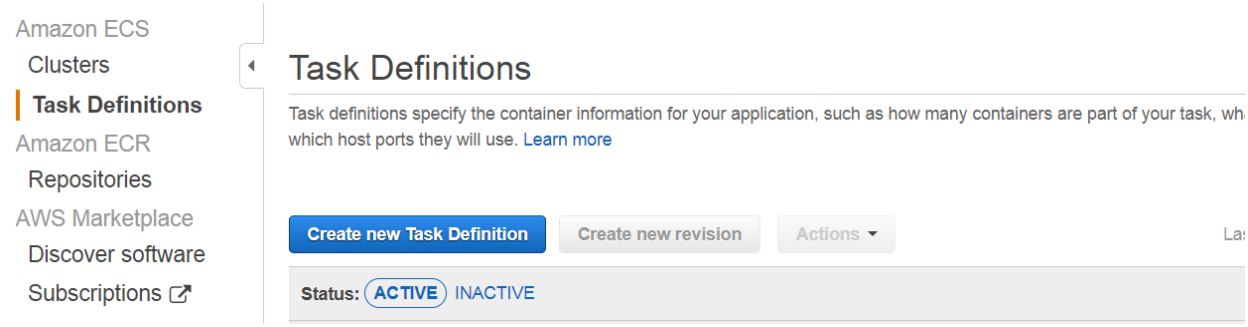
Prerequisites

Make sure there is an repo on the ECR with an container image for which you want to create an container on AWS

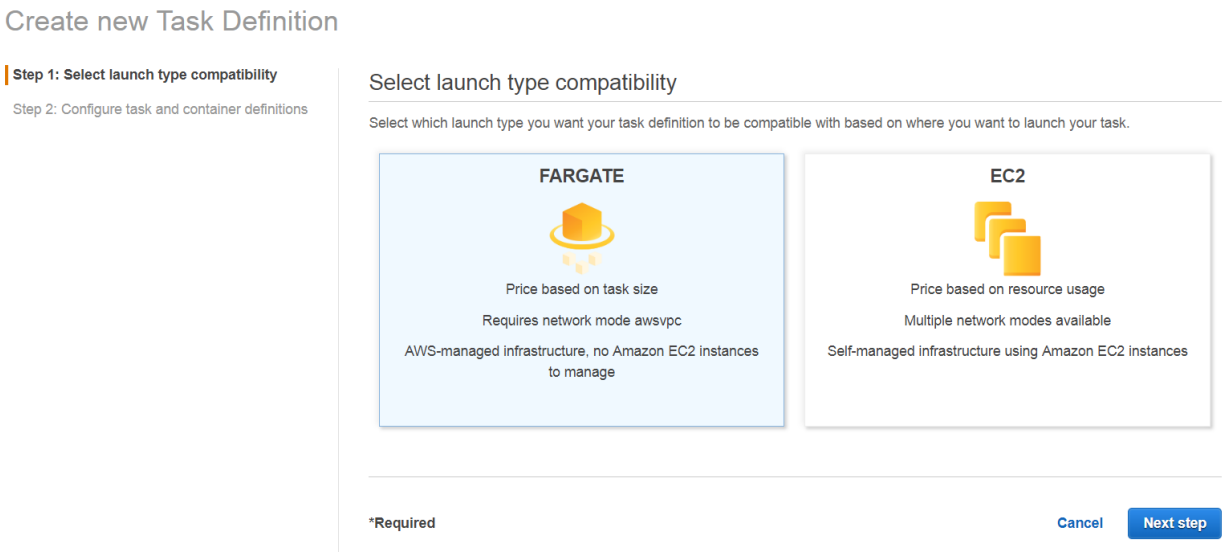
SETUP DETAILS

- Step 1: Create a Task Definition
- Step 2: Configure the Cluster
- Step 3: Configure the Service
- Step 4: Review
- Step 5: (Optional) View your Service
- Step 6: OUTPUT of the Containers

Step 1: Create a Task Definition

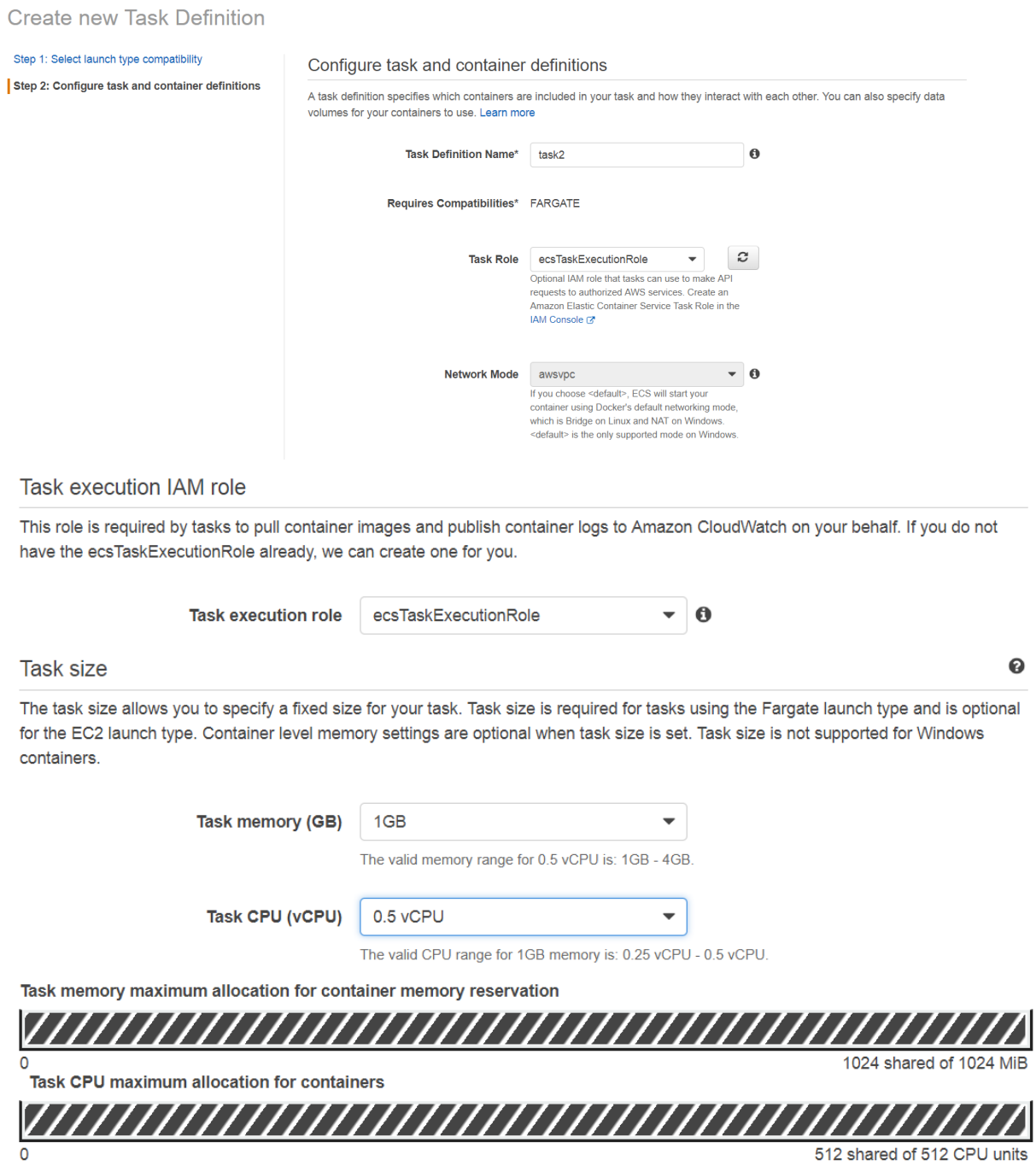


Click on **"Create new Task Definition"**



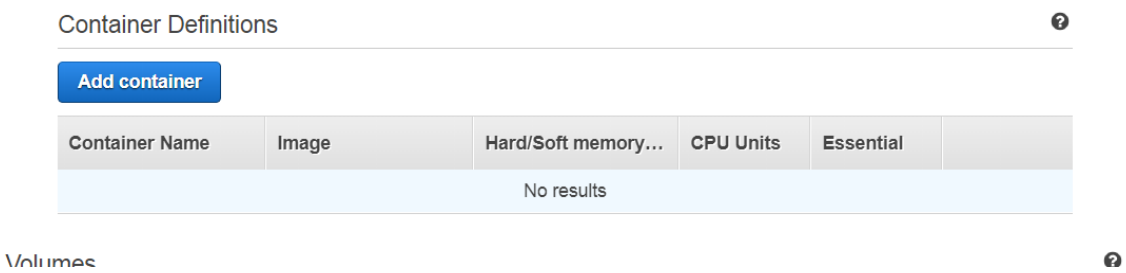
Select "FARGATE" as this Lab Manual will be using "FARGATE" in the service and Cluster creation.

Click **"Next"**

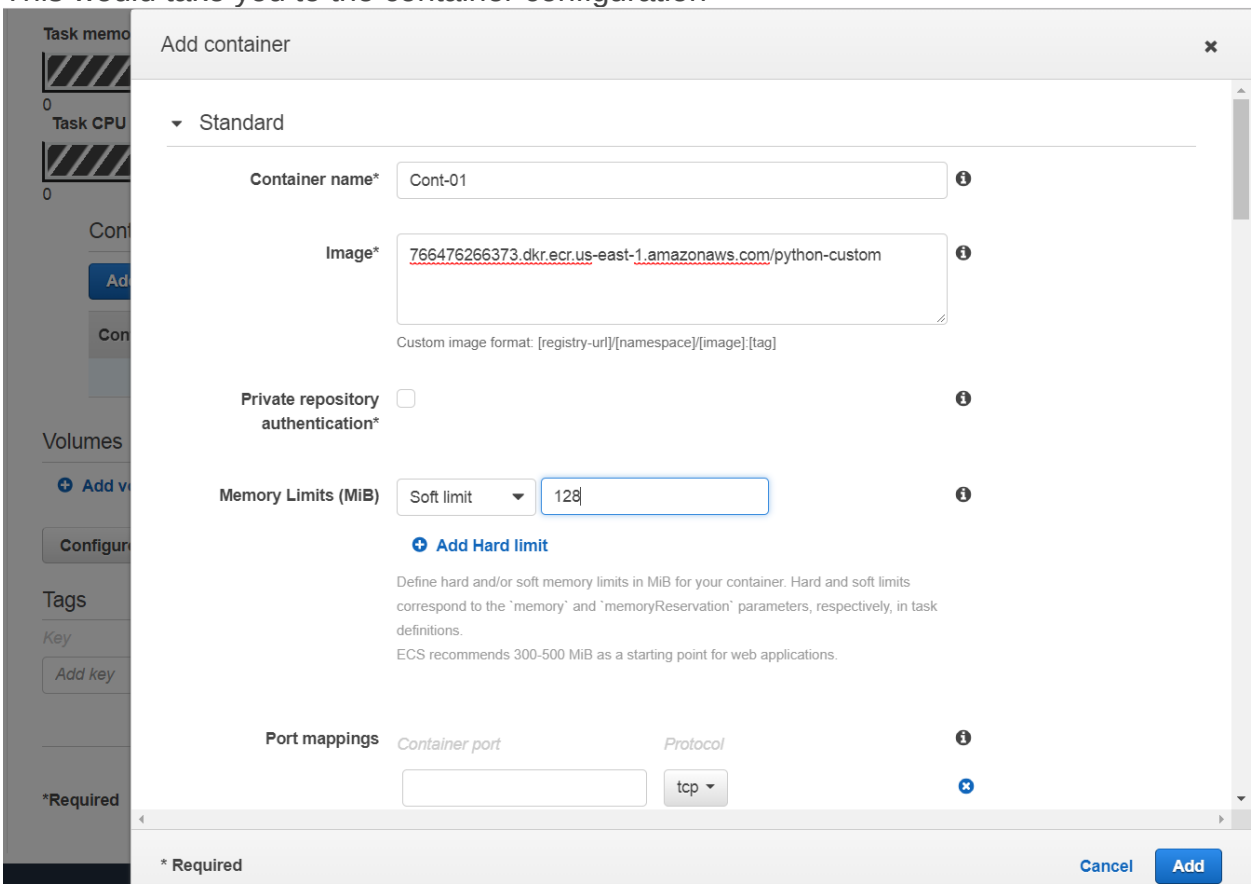


The Task Size is the size limit for all the containers that would be created inside this Task Service.

Next Click on “Add Container”



This would take you to the container configuration



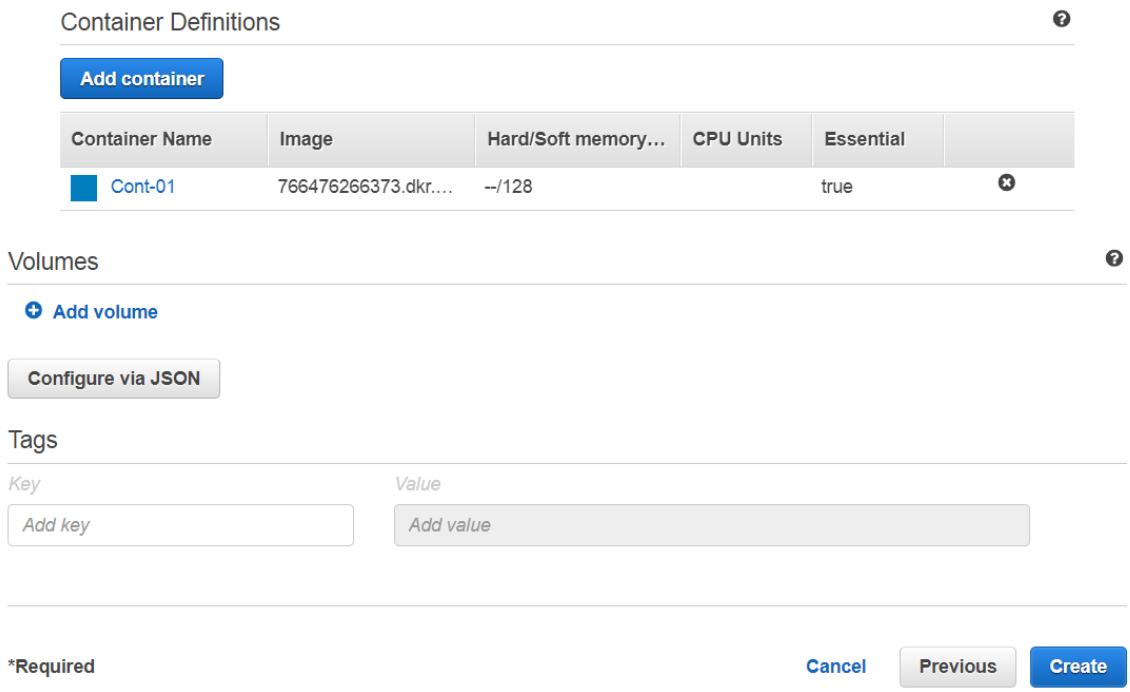
Select the image ID from the **"ECR"**

Set the memory limit (this cannot cross the limit of the Task itself).

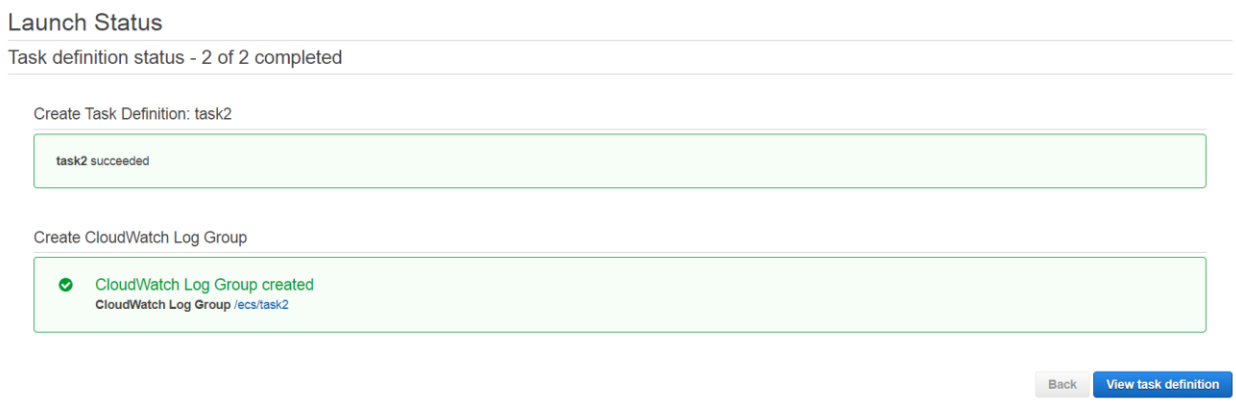
Set the port on which the container would be running the application service.

Example: -- port 80 for our lab,

Click **– "ADD"**

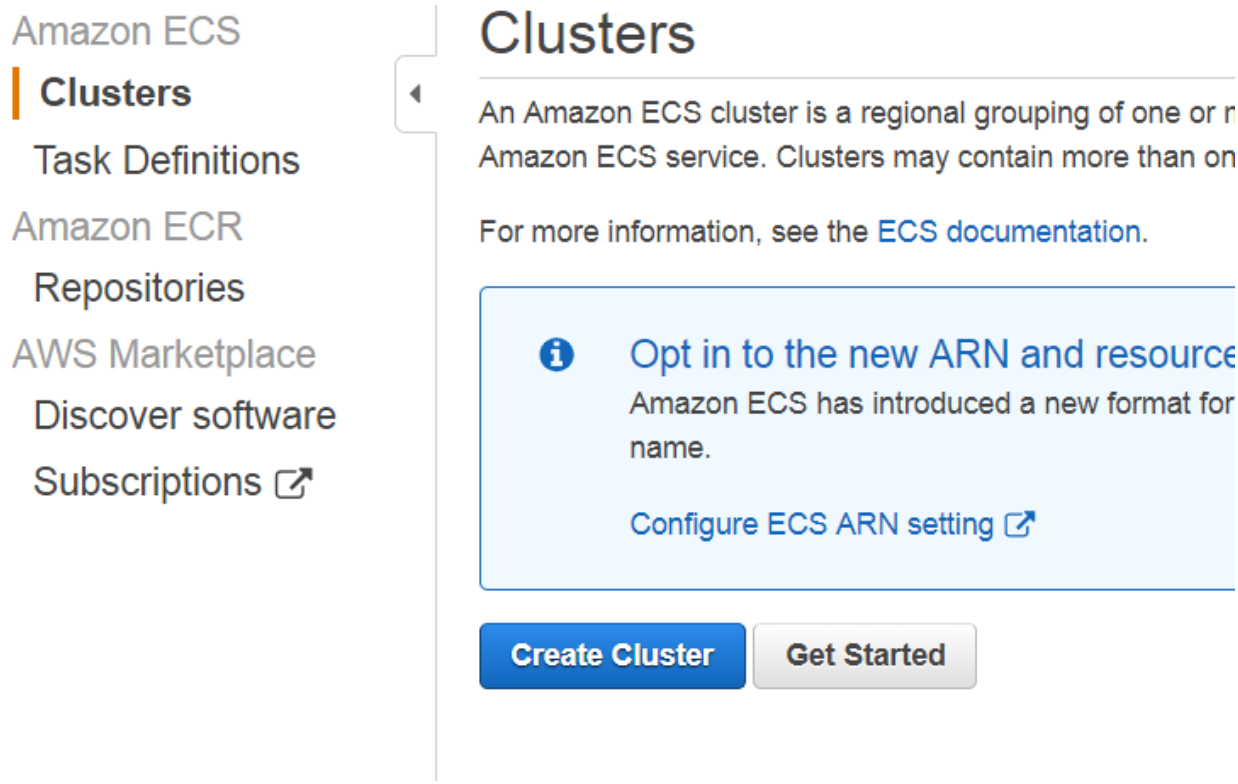


Now Click on "**Create**" to create an Task.

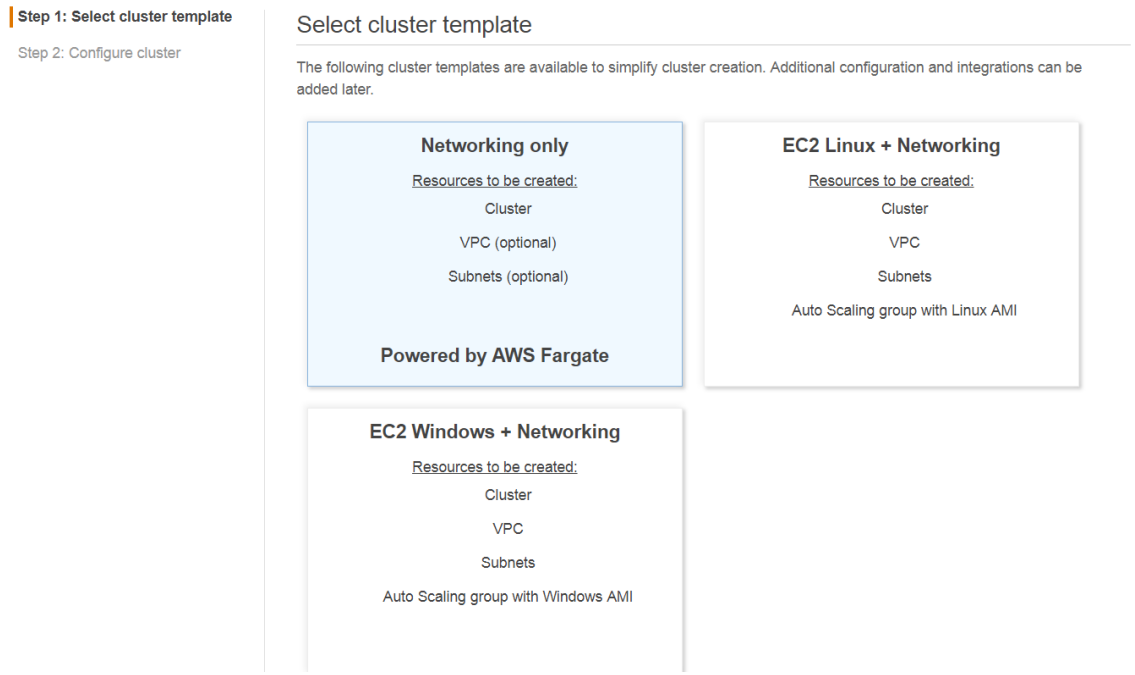


This would be the output of it.

Step 2: Create a Cluster

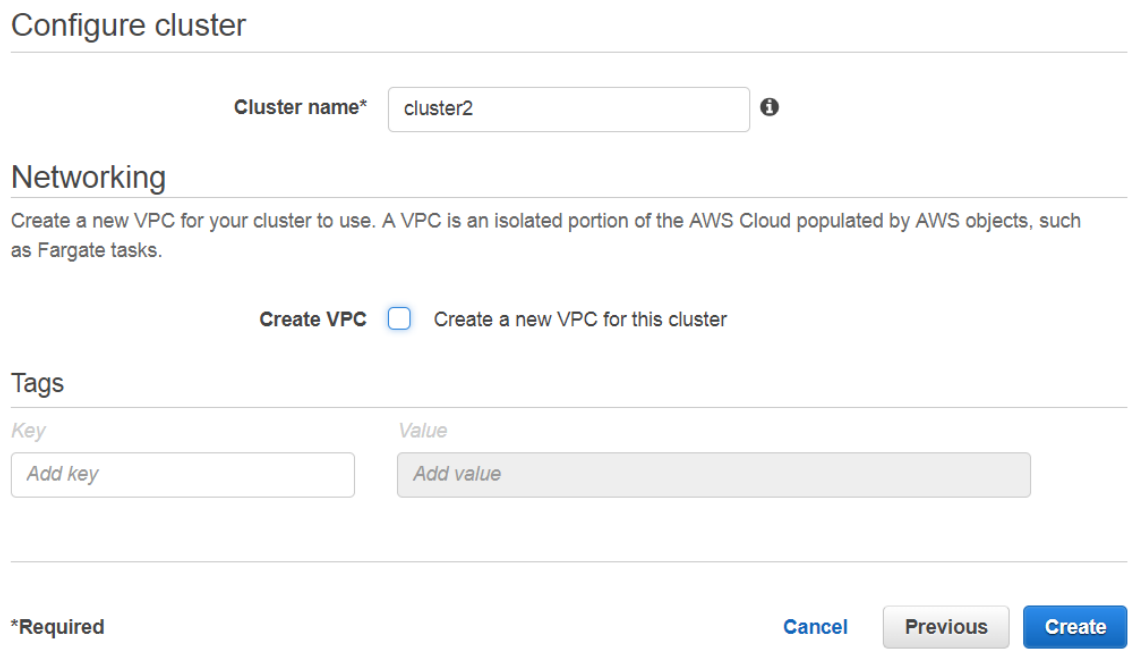


Click on "**Create Cluster**".



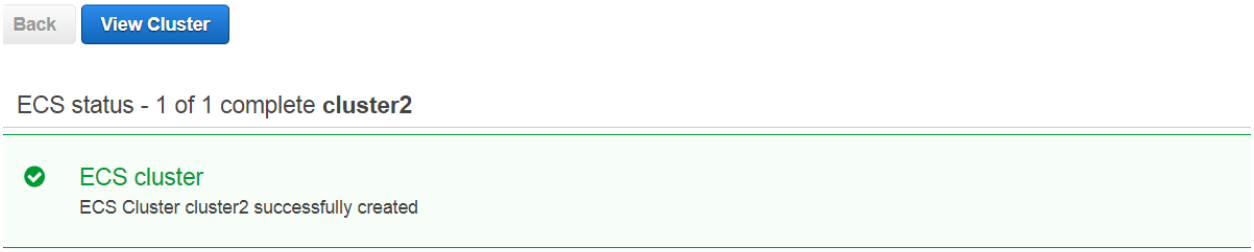
Select “Networking only” for this lab, as we would be using “FARGATE” as done in the Tasks definition.

And Click **“Next Step”**

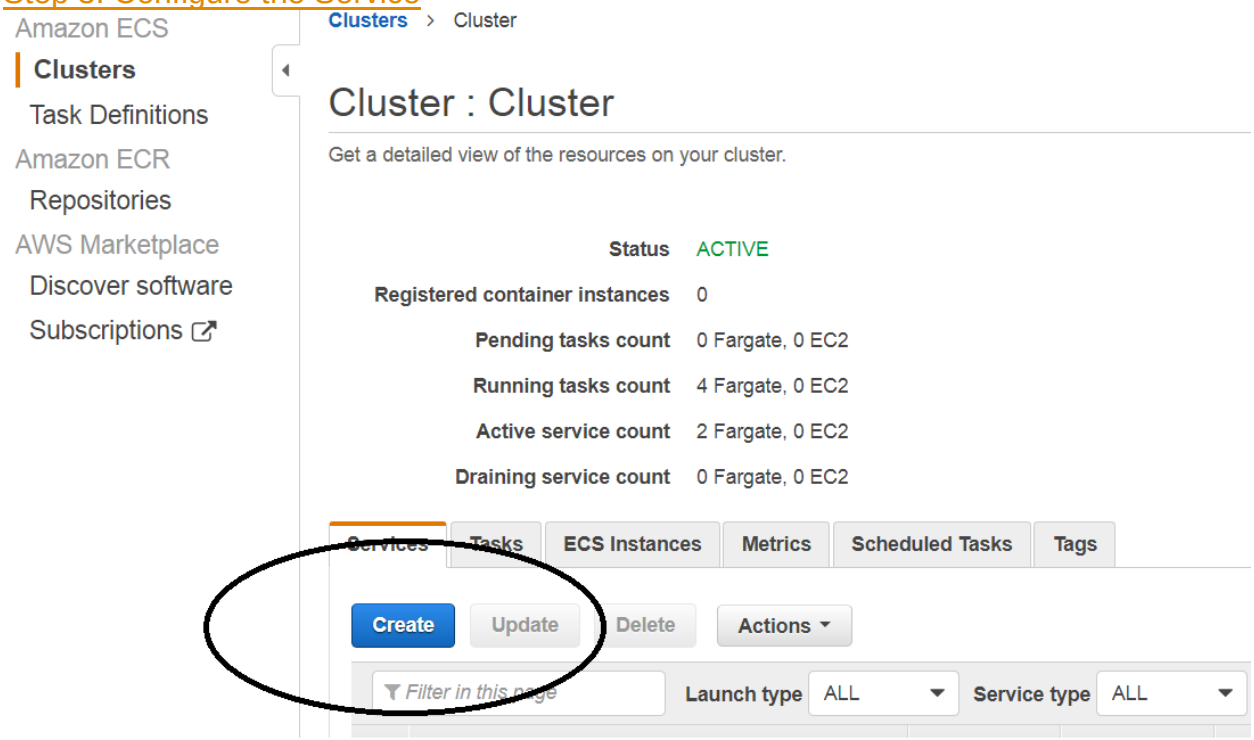


Here either we can create an new VPC or use it in the pool provided by AWS.

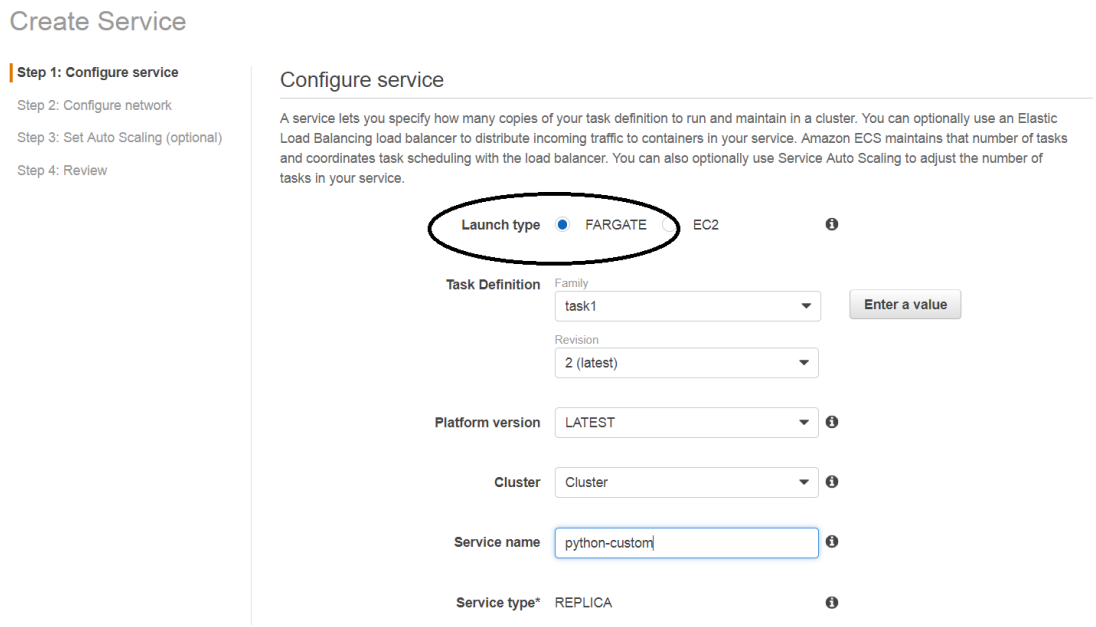
Select **“Create VPC”**
And Click on **“Create”**



Step 3: Configure the Service



Click on “create” to create an Service.



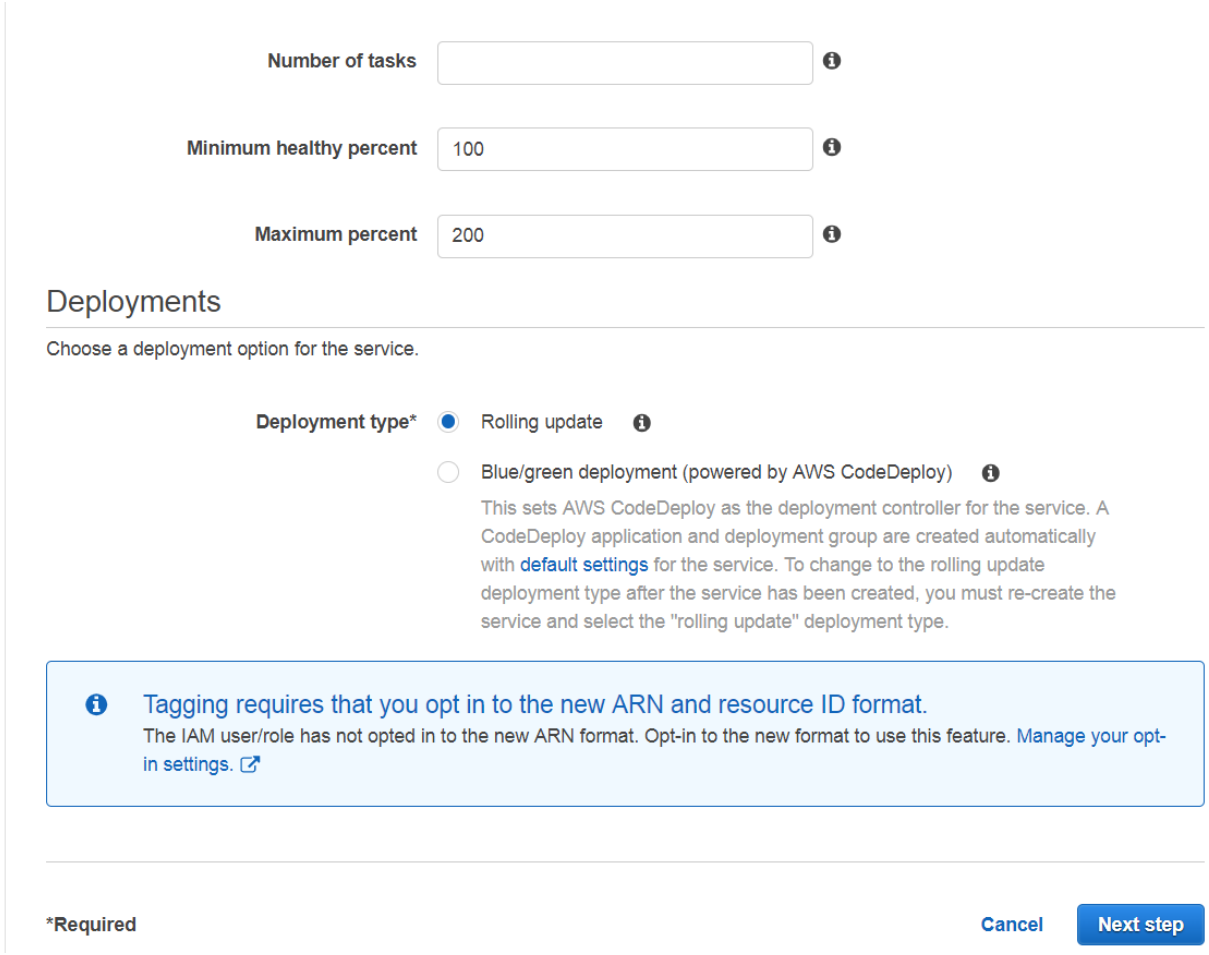
Make you select “FARGATE” on all the 3 places,

Task Definition

Cluster creation

Service creation.

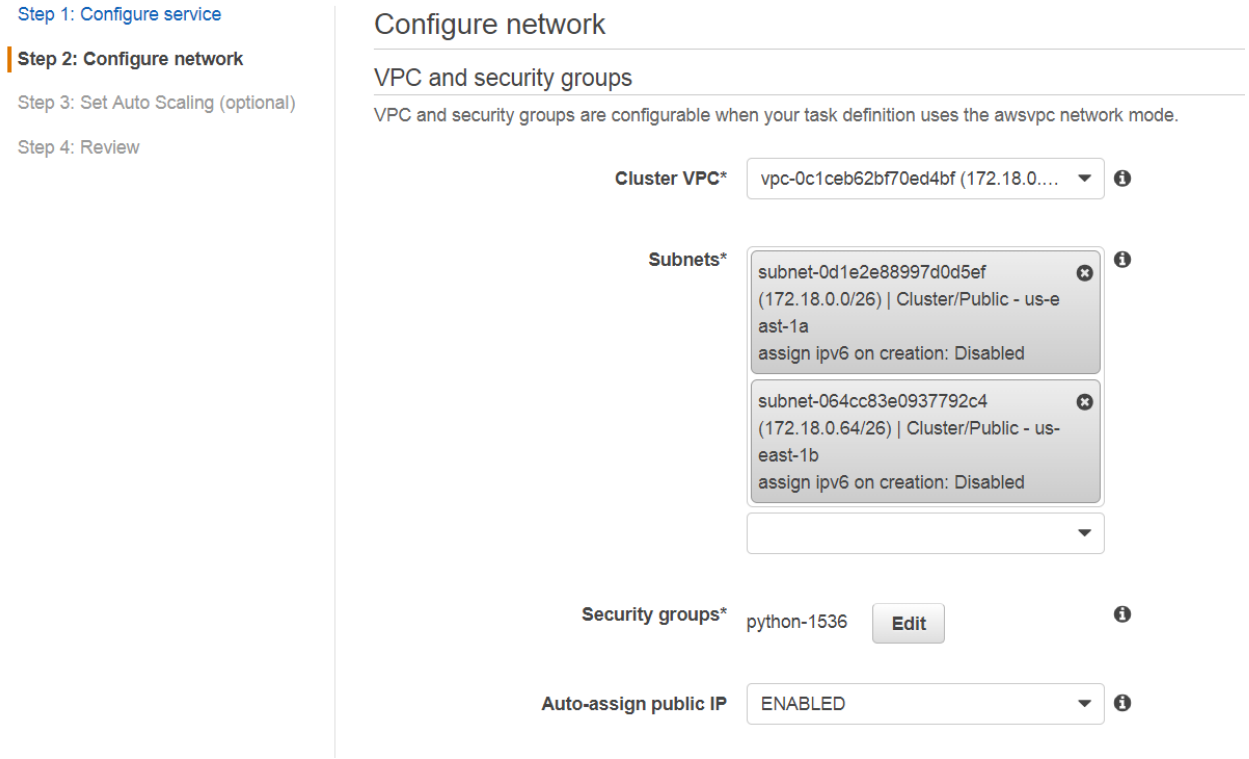
Select the task and the cluster on which you want to run the containers.



In the above, you can mention “HOW MANY TASKS” do you want for a single service. Means if you have 3 containers in a single task and if you say you need 3 TASKS, then totally 9 containers would be created.

I HOPE NO CONFUSIONS. ☺

Click on “Next Step”



Select the VPC and subnets on which you want to run the tasks/containers.

Health check grace period

If your service's tasks take a while to start and respond to ELB health checks, you can specify a health check grace period of up to 7,200 seconds during which the ECS service scheduler will ignore ELB health check status. This grace period can prevent the ECS service scheduler from marking tasks as unhealthy and stopping them before they have time to come up. This is only valid if your service is configured to use a load balancer.

Health check grace period requires a load balancer.

Load balancing

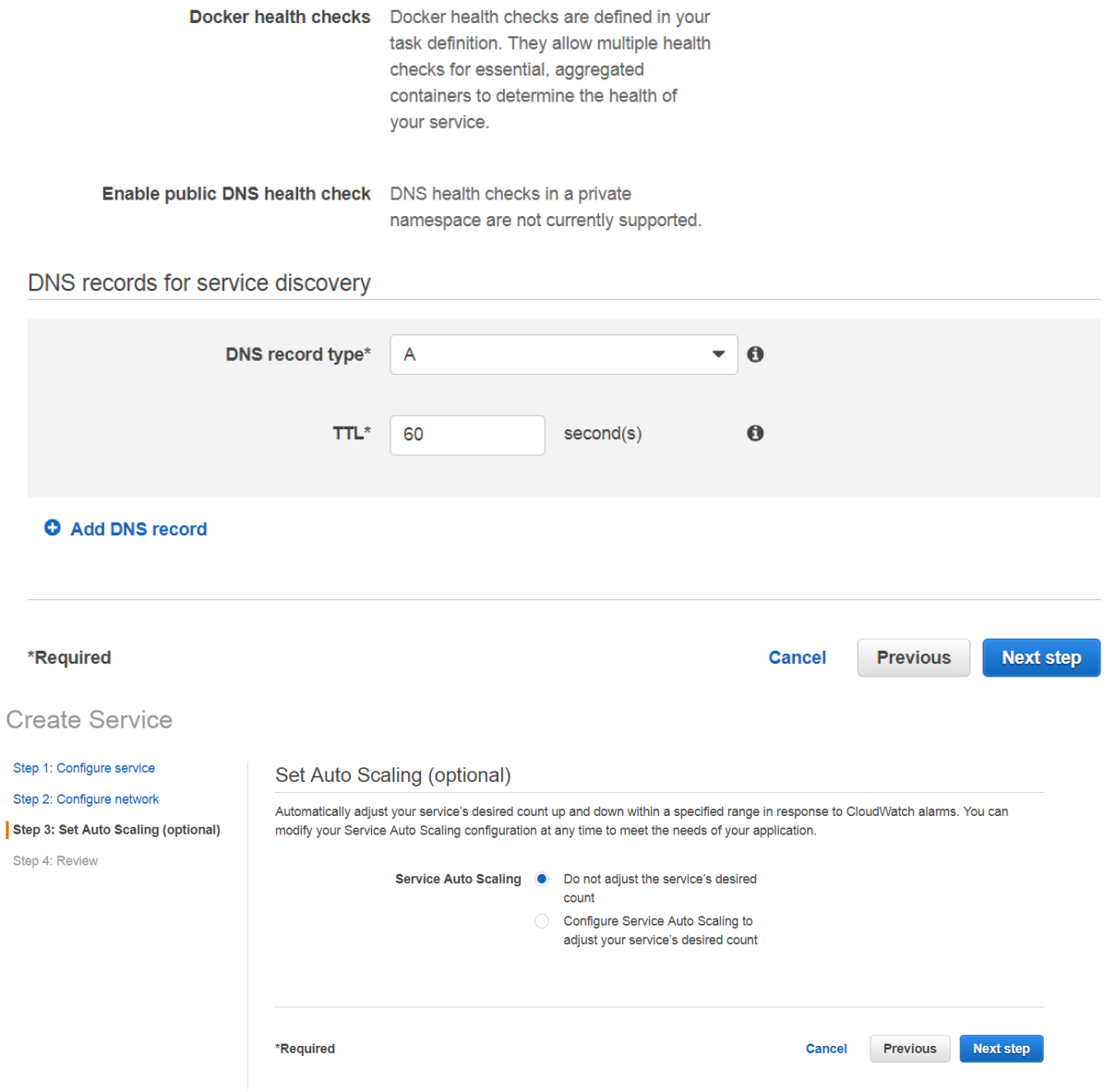
An Elastic Load Balancing load balancer distributes incoming traffic across the tasks running in your service. Choose an existing load balancer, or create a new one in the Amazon EC2 console.

Load balancer type*

- (●) None
  Your service will not use a load balancer.
- ( ) Application Load Balancer
  Allows containers to use dynamic host port mapping (multiple tasks allowed per container instance). Multiple services can use the same listener port on a single load balancer with rule-based routing and paths.
- ( ) Network Load Balancer
  A Network Load Balancer functions at the fourth layer of the Open Systems Interconnection (OSI) model. After the load balancer receives a request, it selects a target from the target group for the default rule using a flow hash routing algorithm.
- ( ) Classic Load Balancer
  Requires static host port mappings (only one task allowed per container instance); rule-based routing and paths are not supported.

For our example “NO LOAD BALANCER”.

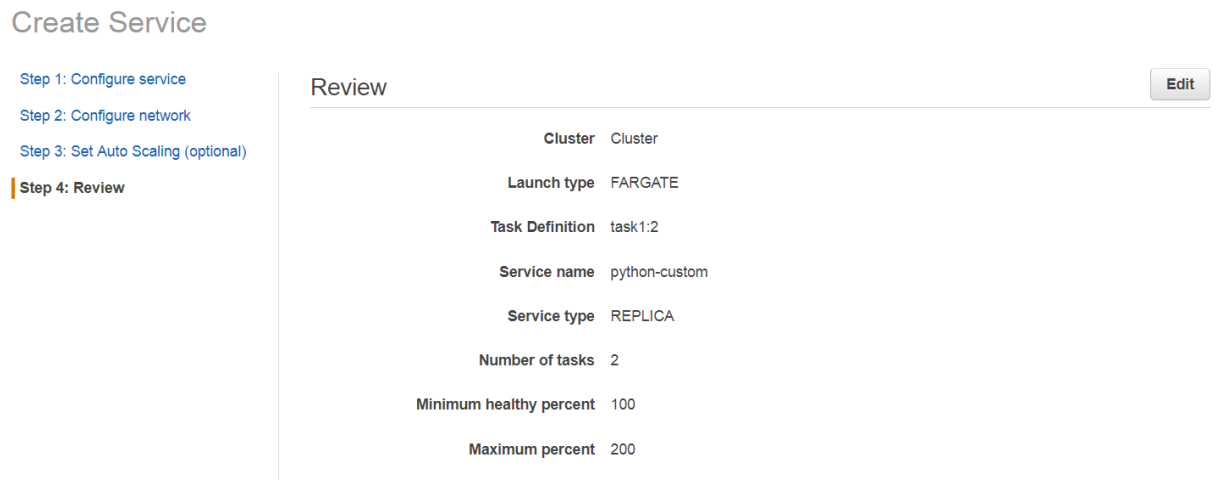
Then click on “Next step” as below



We can set an autoscaling for the number of tasks required to be created if the utilization of the tasks goes high.
Click **"Next Step"**.

Next is to review it.

Step 4: Review



VPC Id	vpc-0c1ceb62bf70ed4bf
Subnets	subnet-0d1e2e88997d0d5ef, subnet-064cc83e0937792c4
Create new security group	python-1536
Auto assign IP	ENABLED

Configure service discovery — Edit

Namespace	ns-vrwxem5efuir2kzf
Service discovery name	python-custom
Enable ECS task health propagation	true
DNS record type and TTL	A 60

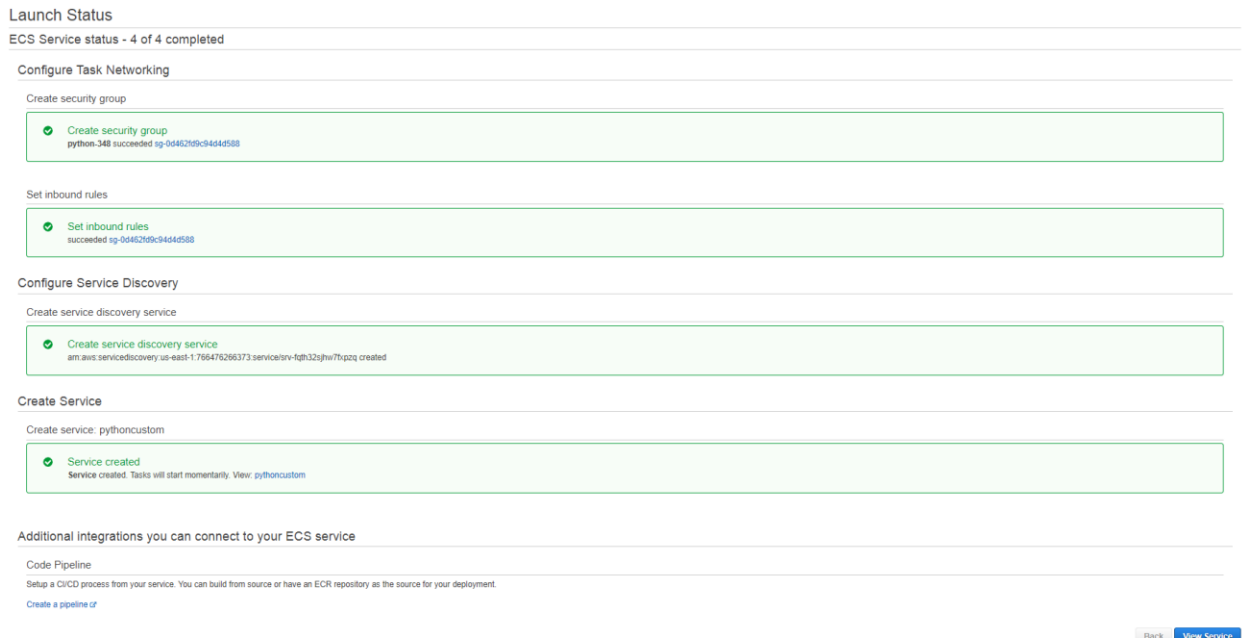
Set Auto Scaling (optional) — Edit

not configured

Cancel | Previous | Create Service

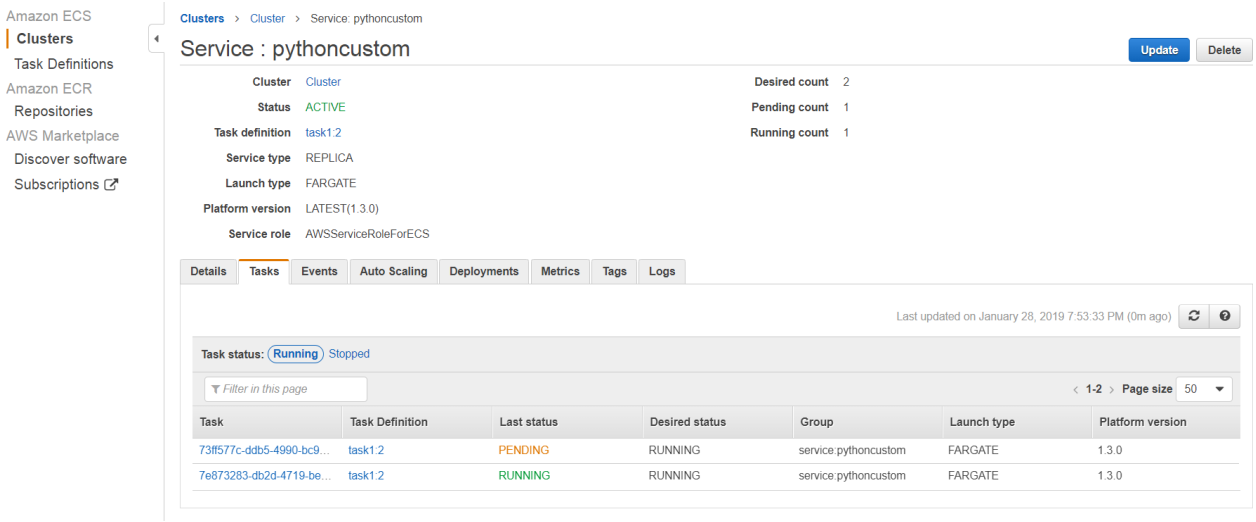
Click on **"Create Service"**

The output should show the below.



Click on "**View service**".

Step 5: View your Service



The above shows that there are 2 TASKS created as per the configuration.

Clusters > Cluster > Task: 73ff577c-ddb5-4990-bc95-7cbb6e00084b

Task : 73ff577c-ddb5-4990-bc95-7cbb6e00084b

Details | Tags | Logs

Cluster	Cluster
Launch type	FARGATE
Platform version	1.3.0
Task definition	task1:2
Group	service:pythoncustom
Task role	ecsTaskExecutionRole
Last status	RUNNING
Desired status	RUNNING
Created at	2019-01-28 19:52:45 +0530
Started at	2019-01-28 19:55:34 +0530

Network

Network mode	awsvpc
ENI Id	eni-0f3cdc0b86fd62edd
Subnet Id	subnet-064cc83e0937792c4
Private IP	172.18.0.126
Public IP	35.153.203.43
Mac address	0e:52:4c:e5:6a:30

Clusters > Cluster > Task: 7e873283-db2d-4719-be44-5732f416f125

Task : 7e873283-db2d-4719-be44-5732f416f12

Details | Tags | Logs

Cluster	Cluster
Launch type	FARGATE
Platform version	1.3.0
Task definition	task1:2
Group	service:pythoncustom
Task role	ecsTaskExecutionRole
Last status	RUNNING
Desired status	RUNNING
Created at	2019-01-28 19:52:45 +0530
Started at	2019-01-28 19:53:22 +0530

Network

Network mode	awsvpc
ENI Id	eni-09f64c472e7be7557
Subnet Id	subnet-0d1e2e88997d0d5ef
Private IP	172.18.0.13
Public IP	34.207.115.9
Mac address	0a:bf:62:b7:a1:be

The above are the details of the tasks.
The private ip and public ip is assigned to the TASKS.

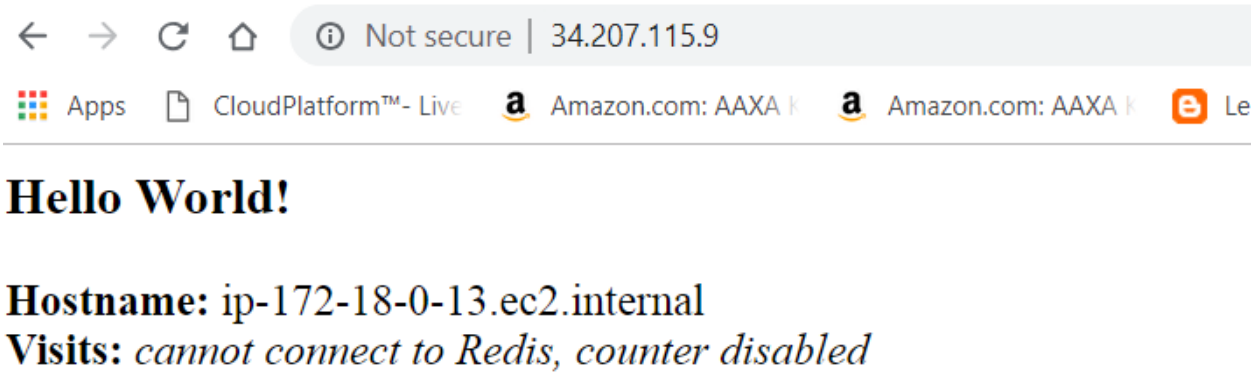
Even if you have multiple containers in the TASKS, all the containers will use the same IP's and different ports if required to be exposed to the external world.

Congratulation's,

YOU HAVE CREATED AN CUSTOM IMAGE CONTAINER ON AWS.

I HOPE YOU ARE STILL WITH ME in this LAB. ☺

Step 6: OUTPUT of the Containers – final step.



This is the same image that we used on the Docker VM /EC2 instance as a custom build. And now it works on the AWS ECS service as well.